

Khushboo K Rao

List of Publications ([ADS link](#))

Published

1. **Rao, K. K.** and Chen, Wen-Ping, “[A Systematic Study of Extended Main-Sequence Turnoffs in Galactic Open Clusters](#)”, 2026, *ApJ*, **1002**, 103
2. Jadhav, V., **Rao, K. K.**, et al., “[A Rotation-based Census of Blue Lurker Candidates in Galactic Open Clusters](#)”, 2026, *A&A*, **707**, A275
3. Dage, K. C., Hunt, E.L., Anderson-Baldwin, Jasmine, Tremou, E., **Rao, K. K.**, et al., “[The Secret Lives of Open Clusters: a Multiwavelength Examination of Three Open Clusters](#)”, 2026, *PASA*, **43**, e013
4. **Rao, K. K.**, Chen, Wen-Ping, “[Investigating Unusually Extended Main Sequence Turnoff of the Galactic Open Cluster NGC 3532](#)”, *IAU Symposium 398 & MODEST-25: Compact Objects and Binaries in Dense Stellar Systems*, June 2025
5. Chand, K., **Rao, K. K.**, Vaidya, K., and Panthi, A., “[Characterization of Blue and Yellow Straggler Stars of Berkeley 39](#)”, 2024, *AJ*, **168**, 278.
6. Balan, S., **Rao, K. K.**, Vaidya, K., Agarwal, M., and Bhattacharya, S., “[Dynamical Evolution of Four Old Galactic Open Clusters Traced by Their Constituent Stars with Gaia DR3](#)”, 2024, *AJ*, **168**, 204.
7. **Kunwar Rao, K.**, Vaidya, K., Agarwal, M., Balan, S., and Bhattacharya, S., “[Blue Straggler Stars: Setting up a Dynamical Clock for Open Clusters](#)”, 2024, *BSRSL*, **93**, 273.
8. Chand, K., **Kunwar Rao, K.**, Vaidya, K., and Panthi, A., “[Hot Stellar Populations of Berkeley 39 Using Swift/UVOT](#)”, 2024, *BSRSL*, **93**, 262.
9. Vaidya, K., Panthi A., Jadhav V., **Kunwar Rao, K.**, Pandey, S., Subramaniam, A., Agarwal, M., “[Blue Straggler Stars of NGC 7789 and NGC 2506 Using AstroSat/UVIT](#)”, 2024, *BSRSL*, **93**, 250.

10. **Rao, K. K.**, Vaidya, K., Agarwal, M., Balan, S., and Bhattacharya, S., “**Determination of dynamical ages of open clusters through the A+ parameter - II**”, 2023, *MNRAS*, **526**, 1057.
11. **Rao, K. K.**, Bhattacharya, S., Vaidya, K., and Agarwal, M., “**Discovery of double BSS sequences in the old Galactic open cluster Berkeley 17**”, 2023, *MNRAS Letters*, **518**, L7.
12. Bhattacharya, S., **Rao, K. K.**, Agarwal, M., Balan, S., and Vaidya, K., “**A Gaia EDR3 search for tidal tails in disintegrating open clusters**”, 2022, *MNRAS*, **517**, 3525.
13. Panthi A., Vaidya K., Jadhav V., **Rao K. K.**, Subramaniam A., Agarwal M., Pandey S., “**UOCS -VIII. UV study of the open cluster NGC 2506 using ASTROSAT**”, 2022, *MNRAS*, **516**, 5318.
14. **Rao, K. K.**, Vaidya, K., Agarwal, M., Panthi, A., Jadhav, V., and Subramaniam, A., “**Characterization of hot populations of Melotte 66 open cluster using Swift/UVOT**”, 2022, *MNRAS*, **516**, 2444.
15. Vaidya K., Panthi A., Agarwal M., Pandey S., **Rao K. K.**, Jadhav V., Subramaniam A., “**UOCS - VII. Blue straggler populations of open cluster NGC 7789 with UVIT/AstroSat**”, 2022, *MNRAS*, **511**, 2274.
16. **Rao, K. K.**, Vaidya, K., Agarwal, M., and Bhattacharya, S., “**Determination of dynamical ages of open clusters through the A+ parameter - I**”, 2021, *MNRAS*, **508**, 4919.
17. Bhattacharya, S., Agarwal, M., **Rao, K. K.**, and Vaidya, K., “**Tidal tails in the disintegrating open cluster NGC 752**”, 2021, *MNRAS*, **505**, 1607.
18. Agarwal, M., **Rao, K. K.**, Vaidya, K., and Bhattacharya, S., “**ML-MOC: Machine Learning (kNN and GMM) based Membership determination for Open Clusters**”, 2021, *MNRAS*, **502**, 2582.
19. Vaidya, K., **Rao, K. K.**, Agarwal, M., and Bhattacharya, S., “**Blue straggler populations of seven open clusters with Gaia DR2**”, 2020, *MNRAS*, **496**, 2402.

Under review:

1. **Rao, K. K.**, Chen Wen-Ping, Chen, Hsiang-Yu, Dage Kristen, and Offner Stella S. R., “**Uncovering the Origin of Slow Rotators in the Young Star-Forming Cluster Trumpler 14**”, submitted to *APJ* on 27-Mar-2026 (reference: AAS75459)
2. **Rao, K. K.** and Banerjee, Sambaran, “**Examining the stellar-merger origin of the blue main sequence in the open cluster NGC\,3532 with N-body simulations**”, 2026, *submitted to A&A* (reference: aa62082-26)

In Pipeline:

3. **Rao, K. K.**, Jadhav, V., et al., “**Straggler Stars through Variability – II. Light Curve Modelling of Eclipsing Blue Straggler Stars**”, in an advanced stage
4. Jadhav, V., **Rao, K. K.**, “**Straggler Stars through Variability – I. Photometric Parameters of Eclipsing and Pulsating Blue Straggler Stars**”, in an advanced stage
5. **Rao, K. K.**, et al., “**Hints of Angular Momentum Loss in Intermediate-Mass Stars due to High-Energy Activities in NGC 3293**”, in a preliminary stage
6. **Rao, K. K.**, et al., “**Variability Study of Massive Members of NGC 6940 Open Cluster using High Cadence TAOS II Photometric Data**”, in a preliminary stage